

ROOPANSHU GUPTA

Research engineer: experiment loops, evaluation design, and LLM agents

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EDUCATION

Delhi Technological University — B.Tech in Software Engineering Expected May 2028
CGPA: 9.67 / 10 (Departmental Rank 1, Semester 2) | Class XII 93.8% | Class X 97.33%

EXPERIENCE

Machine Learning Research Intern | Machine Learning Research Lab, DTU Jan 2026 – Present

- Ran a **hypothesis → experiment → ablation loop** on CycleGAN and conditional diffusion models for SAR-to-optical satellite translation, taking **PSNR from 15 to 31.58 dB** and **SSIM from 0.18 to 0.87** through attention, custom losses (Charbonnier, MS-SSIM, perceptual) and ControlNet conditioning.
- Built the **experiment infrastructure** behind it: a 1TB+ satellite-imagery pipeline (acquisition, pairing, preprocessing) and S3/HPC prefetching that removed I/O stalls across concurrent multi-GPU jobs.

5G AI Systems Intern | DoT 5G Use Case Lab, DTU Jun 2025 – Jul 2025

- Fine-tuned a **YOLOv8** traffic-analytics model to **97.5% mAP@0.50** and 90.0% precision; applied **FP16 quantization** to cut inference latency for real-time deployment on 5G edge hardware.

PROJECTS

trusted-evolve: Verified Evolutionary Search on AlphaEvolve's Circle Packing | [GitHub](#)

Python, NumPy/SciPy (SLSQP, analytic Jacobians), multiprocessing, Claude API (structured outputs)

- Evolution + SLSQP local search under a **zero-tolerance verifier** reached **2.635983** (n=26, sum of radii): above AlphaEvolve's 2.635863 and **equal to the best known packing**, in ~1 min per run on a laptop CPU; best variant hit it in **6/10 seeds** (random restarts: 1/10).
- A 30-run, 10-seed study found the **swap operator hurts** (p = 0.004) and that evolution vs restarts is **not significant** (p = 0.26): a 1-seed comparison picks the wrong winner **30%** of the time. Built a Claude-proposed experiment loop that accepts changes only via a code-enforced Welch-test gate.

Cloud Remediation Agent + RemediationBench | [GitHub](#)

Python, Claude API (tool use), Model Context Protocol, boto3, moto, pytest

- Agent benchmark of 7 sandboxed AWS incidents (incl. red herrings) graded by **hidden state verifiers**; a code-enforced verify stage cut false-success **29% → 0%**, and verifiers caught 4 damaging actions that no static policy rule flagged.

CAF-OTSRNet: Optical-Guided Thermal Super-Resolution | [GitHub](#)

PyTorch, cross-attention, deformable alignment

- Cross-attention fusion network reconstructing high-resolution thermal imagery from low-res thermal + high-res optical inputs: **31 dB PSNR, 0.85 SSIM on Landsat-8**.

Unified Data MCP Server | [GitHub](#)

Python, MCP, sqlalchemy, JWT / OAuth 2.1

- Read-only, audited agent access to SQL + REST; AST-level SQL guard **blocked 48/48 red-team attacks** with 0/18 false positives (92 tests).

ACHIEVEMENTS

- **All India Rank 104**, Indian Statistical Institute entrance; **JEE Main AIR 7997** (99.5 percentile).
- **First Prize**, DTU Summer School on AI (with Adobe) | **National Finalist, Smart India Hackathon 2025** (led team, ISRO problem statement) | **Top 53 / 2,000+**, NatWest Code for Purpose.

SKILLS

Languages: Python, C/C++, Java, SQL

Research & ML: PyTorch, diffusion models, GANs, computer vision, numerical optimisation (SciPy), experiment design, ablations, statistical testing

Agents & Systems: Claude API (tool use, structured outputs), MCP, agent evaluation, Docker, FastAPI, AWS, Linux, HPC / multi-GPU, Git, pytest